

Tethered 6U CubeSat

Tether range, relative attitude, state of the art and 3D imaging

CUBICS 2026 · Stream 1 · Team meeting, October 2026

One tether, two 3U halves of 4 kg, circular orbit at 550 km.

Four answers for today

Tether range	Vertical only. 500 m to 1 km is feasible: 3.6 to 7.2 mN of tension, about 3.5 h to deploy.
Relative attitude	IMUs alone drift. A star tracker and a GNSS receiver on each half; the stowed zero only calibrates.
State of the art	Tethers have flown since 1966. Most failures happen during deployment.
3D photos	Not with a 0.5 to 1 km tether baseline: height errors near 1 km. Yes with fore/aft camera pointing: about 3.5 m.
Lifetime	Reel the tether in when not in use: about 6 years instead of 1.6 years at 550 km.

The tether is held taut by the gravity gradient

Circular orbit of radius $a = R_{\oplus} + h$:

$$n = \sqrt{\frac{\mu}{a^3}} = 1.0948 \times 10^{-3} \text{ rad/s}, \quad P = \frac{2\pi}{n} = 95.6 \text{ min.}$$

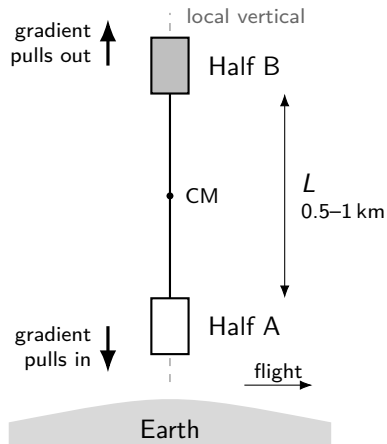
A body forced to turn at rate n at radius $a + l$ lacks

$$n^2(a + l) - \frac{\mu}{(a + l)^2} \simeq 3 n^2 l$$

of inward acceleration. The tether supplies it:

$$T = 3 n^2 m_{\text{eq}} L, \quad m_{\text{eq}} = \frac{m_1 m_2}{m_1 + m_2} = 2 \text{ kg.}$$

$$T = 3.6 \text{ mN (500 m)} \quad T = 7.2 \text{ mN (1 km)}$$



Only the vertical direction separates the two halves

Relative motion in the orbiting frame (x radial, y along track, z out of plane):

$$\ddot{x} - 2n\dot{y} - 3n^2x = f_x, \quad \ddot{y} + 2n\dot{x} = f_y, \quad \ddot{z} + n^2z = f_z.$$

Direction	Free motion	Tether
Radial (x)	$+3n^2x$: the halves move apart	taut
Along track (y)	no restoring or separating term	slack
Out of plane (z)	$-n^2z$: the halves move together	slack

A tilt φ from the vertical swings back (constant L):

$$\ddot{\varphi} + 3n^2 \sin \varphi \cos \varphi = 0 \Rightarrow \omega_{\text{lib}} = \sqrt{3} n, \quad 2\pi/\omega_{\text{lib}} = 55.2 \text{ min.}$$

A diagonal tether is a swing, not a configuration. Holding $\varphi = 30^\circ$ at 1 km needs $3n^2 m_{\text{eq}} L^2 \sin \varphi \cos \varphi = 3.1 \text{ N m}$, about 3000 times a CubeSat reaction wheel.

500 m to 1 km with one tether: the key numbers

Quantity	Expression	$L = 500 \text{ m}$	$L = 1 \text{ km}$
Tension	$3n^2 m_{\text{eq}} L$	3.6 mN	7.2 mN
Elastic stretch	$TL/(EA)$	0.6 mm	2.3 mm
Thermal length change, $\Delta\theta = 100 \text{ K}$	$ \alpha L \Delta\theta$	0.15 to 0.5 m	0.3 to 1.0 m
Tether mass, $\varnothing 0.2 \text{ mm}$	$\rho_c AL$	22 g	44 g
Tether frontal area	$d L$	0.10 m^2	0.20 m^2
Controlled deployment, $\lambda = n/2$	$\ln(L/L_0)/\lambda$	3.2 h	3.5 h
Controlled retrieval, $\lambda = n/5$	$\ln(L/L_0)/\lambda$	7.9 h	8.8 h
Orbit life at 550 km, deployed	drag model	$\approx 2.5 \text{ yr}$	$\approx 1.6 \text{ yr}$

$EA = 3142 \text{ N}$, $\alpha \approx -10^{-5}/\text{K}$ (polyethylene) to $-3 \times 10^{-6}/\text{K}$ (aramid), $L_0 = 1 \text{ m}$, mean solar activity.

Stowed: $\approx 6 \text{ yr}$.

Deployment has to be slow

In-plane dynamics with tether length L and tilt φ :

$$\ddot{L} = L[(\dot{\varphi} + n)^2 - n^2(1 - 3\cos^2 \varphi)] - \frac{T}{m_{\text{eq}}},$$

$$\ddot{\varphi} = -2\frac{\dot{L}}{L}(\dot{\varphi} + n) - 3n^2 \sin \varphi \cos \varphi.$$

The term $-2\dot{L}/L$ (Coriolis) tilts the tether while it pays out.

Free, no brake ($T = 0$, $\varphi \equiv 0$):

$$L(t) = L_0 \cosh(\sqrt{3} n t)$$

75 to 81 min, but ends at 0.95 to 1.9 m/s and the tether tips over.

Controlled, $L = L_0 e^{\lambda t}$:

$$\sin 2\varphi^* = -\frac{4\lambda}{3n}, \quad \lambda \leq \frac{3}{4} n.$$

With $\lambda = n/2$: $\varphi^* = -20.9^\circ$ and 3.2 to 3.5 h.

Once deployed, the length stays fixed

The winch sets L . More tension does not lengthen the tether; it only pulls it tighter.

Elastic stretch

$$\Delta L_{\text{el}} = \frac{T L}{EA}$$

$$\leq \mathbf{2.3 \text{ mm}}$$

at 1 km and full tension

Thermal change

$$\Delta L_{\text{th}} = |\alpha| L \Delta\theta$$

up to 1 m

sun to shadow; the winch
compensates

Natural swing

$$2\pi/(\sqrt{3} n)$$

55 min

period around the vertical

The tension is fixed by the length: $T = 3n^2 m_{\text{eq}} L$. Paying out faster only adds swing and braking load.

Relative attitude: IMUs alone are not enough

Why the stowed zero is lost

- Gyros integrate the rate, so a bias b grows: $\delta\theta(t) \simeq b t$.
- $b = 1^\circ/\text{h}$ gives 1.6° after one orbit.
- Accelerometers cannot sense attitude in free fall.
- One tether does not stop the halves from turning about its own axis.

Stowed zero: calibrates the mounting offset between the halves only.

What to fly on each half

- **Star tracker:** absolute attitude, $\approx 0.005^\circ$.

$$R_{AB} = R_A^\top R_B$$

- **GNSS:** tether direction from the baseline

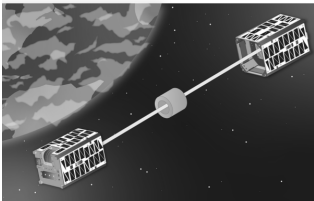
$$\hat{u} = \frac{\mathbf{r}_B - \mathbf{r}_A}{\|\mathbf{r}_B - \mathbf{r}_A\|}, \quad \delta\hat{u} \approx \frac{\delta r}{L} \approx 10^{-5} \text{ rad}$$

- **Tether encoder** at each anchor; **IMU** for fast rates.

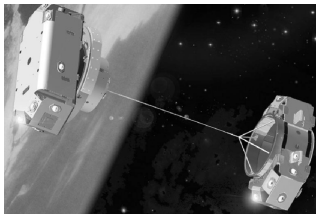
Tether heritage: deployment is the main risk

Mission	Year	Length	Result
Gemini XI	1966	30 m	First tether between two spacecraft
SEDS-1	1993	20 km	Full deployment; end mass deorbited
SEDS-2	1994	19.7 km	Closed-loop braking law validated
TSS-1R	1996	19.7 km	Tether broke by an electrical arc
TiPS	1996	4 km	Libration damped from 40° to 7.5°
ATE _x	1999	22 m of 6.2 km	Tether went slack after a thermal transient
YES2	2007	31.7 km	Longest structure deployed in orbit
TEPCE	2019	1 km	Two CubeSat halves split by a stacer spring at 4 m/s

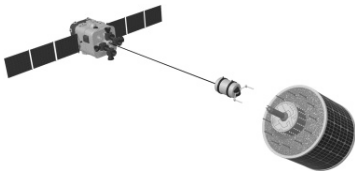
Previous tethered systems



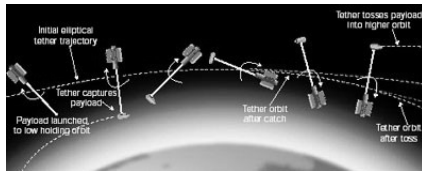
Two CubeSats joined by a tether



Two end bodies joined by a tether



Tether between a spacecraft and a target



Momentum exchange: catch and toss

Where our proposal fits

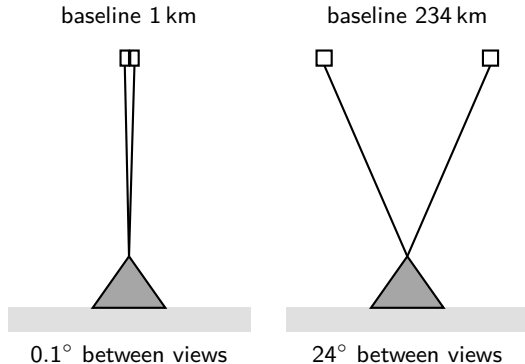
Already demonstrated

- Splitting a CubeSat in two with a spring (TEPCE, 2019)
- Kilometre-class deployment with braking (SEDS-2, TiPS)
- Gravity-gradient stabilisation along the vertical
- Libration damping with the winch

New in this proposal

- Active thermal compensation of the tether length (the ATEx failure)
- Tether direction measured at the anchor
- Tether reeled in between campaigns to extend the mission
- Earth imaging from both halves

3D needs two very different viewing angles



A point of height h shifts between the photos by

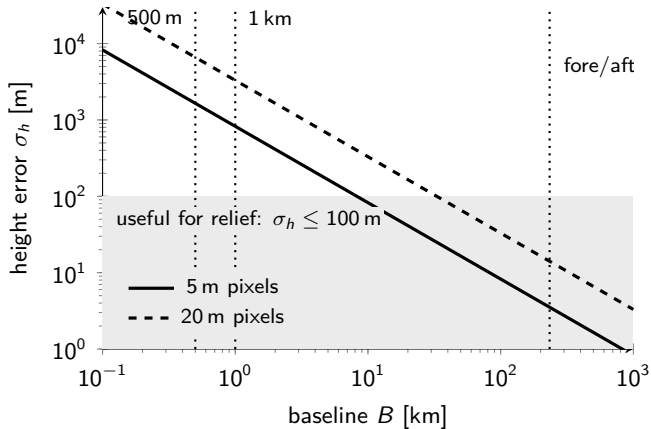
$$p = h \frac{B}{H},$$

so the height error is

$$\sigma_h = \frac{\sigma_p}{B/H}.$$

B baseline, $H = 550$ km, σ_p matching error (0.3 pixel = 1.5 m with 5 m pixels).

A 0.5 to 1 km baseline gives height errors near 1 km



Tether baseline, even if horizontal: $B/H = L/H$.

- 1 km: $\sigma_h \approx$ **825 m**
- 500 m: $\sigma_h \approx$ **1.65 km**
- A 1 km mountain shifts **0.36 pixel**.

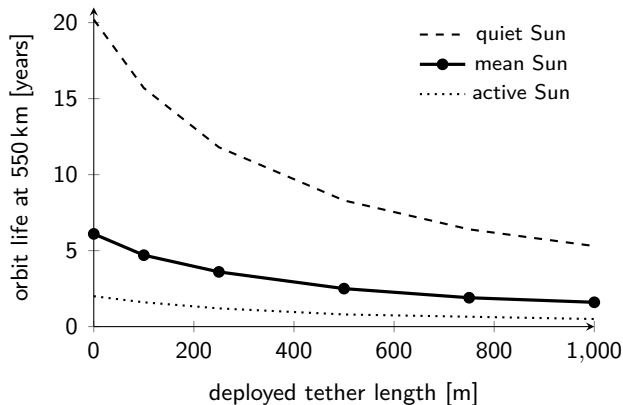
A vertical tether gives no stereo at all: both cameras look along the same line.

What works: pointing the cameras

Option	B/H	σ_h (5 m / 20 m pixels)	Notes
Fixed, A $+12^\circ$ / B -12°	0.43	3.5 m / 14 m	Sites under the track
Gimbal $\pm 30^\circ$	0.40	3.8 m / 15 m	Chosen sites, 2–4 days
Tether baseline, 1 km	1.8×10^{-3}	825 m / 3.3 km	Not usable

The 3D comes from the satellite moving 234 km along its orbit in 31 s, not from the tether.

Keep the tether stowed when it is not in use



Drag decay of the orbit radius:

$$\dot{a} = -\rho C_D \frac{A}{m} \sqrt{\mu a}, \quad A = A_{\text{sat}} + d L.$$

A 1 km tether of 0.2 mm adds 0.20 m^2 , three times the satellites.

- **Reel in between campaigns:** about 6 yr instead of 1.6 yr.
- **Deploy at end of life** to reenter faster: at 590 km, 3 yr instead of 11.5 yr.

Other risks

Retrieval	Reeling in amplifies the swing ($-2\dot{L}/L > 0$). It needs libration control and a slow rate: 7.9 to 8.8 h.
Thermal length	Up to 1 m per orbit at 1 km. The winch needs that range and closed-loop control.
Single tether	One cut by a small debris particle ends the mission. Use a braided or multi-strand line.
Debris and admissibility	A 1 km object has a larger collision cross-section. Confirm with the CSA.
Model accuracy	Orbit lives come from a simple drag model; a formal STELA or DRAMA study is needed.

Decisions for the team

- 1 **Tether length:** 500 m or 1 km, with the tether stowed between campaigns.
- 2 **Attitude sensors:** star tracker and GNSS on each half; IMU and encoder as support.
- 3 **Imaging claim:** drop tether stereo; use fore/aft or gimbal cameras for 3D.
- 4 **Questions to the CSA by 9 November:** launch altitude, tether admissibility, deorbit case.
- 5 **Analysis:** formal orbit-life study and a thermal model of the tether.